

State diversity in different neural network structures

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Abstract: The objective of this work is to study the capability to replicate different states of some different topological configurations named as *feed forward*, *sunrise* and *circular* with different inter-module connections. To achieve this, the Izhikevich mathematical model has been used for excitatory regular spiking neurons. A sort of simulations has been done with different values of the noise intensity for a total of 7 different configurations, in order to analyze the evolution of the functional complexity ϕ_{FC} and to find the value of the maximum with an adjustment. Finally, the maximum values have been compared and it has been identified that the best configuration to replicate different states of the neural network is the *feed forward* with all the inter-modular connections in the same direction, showing the important effect of directionality.

Keywords: Numerical methods, complex systems, statistical physics

SDGs: Health & Wellness

I. INTRODUCTION

The human brain is among the most complex systems in nature, composed of billions of interconnected neurons that give rise to behavior, cognition, and consciousness. In neural science, there has been considerable progress modeling an individual neuron and its behavior, but it is still a challenge to study the dynamics of interacting neural networks at larger scales. It is hardly impossible to do so or to predict just by observing neurons. That prompted the elaboration of mathematical models that, with the advances in computing, can simulate models of interacting neurons.

The dynamics of a neural network are interesting to study how certain patterns can affect to human brain, for example synchronization of different networks. It is known that a strong synchronization can be a sign of abnormal behavior, which can lead to diseases or neurological conditions such as epilepsy or schizophrenia. On the other hand, the lack of it can be beneficial and enhance positive aspects such as memory or attention. The purpose of studying it is not only the fact that neurons can be synchronized, because of their simultaneous firings, but also seeing the regular patterns that appear, the collective bursting or the dynamics of a neural system.

The objective of this work is to study different topological configurations and to verify that exists a structure that is the most capable of replicating the major number of different states. To achieve this, different simulations have been done for every system varying the intensity of noise the system receives and only excitatory regular spiking (RS) neurons have been considered. This work does not emphasize in which are those states or under which conditions of noise it gets maximum, as noise is a parameter that can not be defined in experiments with real neurons.

In this study, the model that has been used is the Izhikevich model. This is a mathematical model that has

the advantage of having a low computational cost and reproduces realistic dynamics of a neural network, making it easier to study large-scale neural networks. This model allows the study of different phenomena that can appear in a neural network, such as the mentioned before, and is able to replicate the behavior of different types of neurons.

The results of a single simulation with a fixed noise intensity will be a list of firings of each neuron at every time step taking 1 or 0 (True/False) if they fired at a certain step or not, respectively. After the data analysis, the capability of the system to replicate different states will be quantified in the calculation of the functional complexity. This parameter is bounded between 0 and 1 and takes 1 when the system is able to replicate every possible state.

Finally, the result for each topological configuration will be the evolution of the functional complexity for different noise intensities. The final result will be the comparison of the maximum value of the functional complexity for the different cases. To achieve this an adjustment will be done for every case.

II. METHODOLOGY

A. Mathematical neuron model

The neurons in the spiking neural network (SNN) were modeled using Izhikevich model. It consists in a two-variable system with two ordinary differential equations, for each neuron. This means that in a system of N neurons there will be $2N$ equations and $2N$ independent variables. The form of these equations is

$$\begin{cases} \dot{v}_i = 0.04v_i^2 + 5v_i + 140 - u_i + \eta + I_i \\ \dot{u}_i = a(bv_i - u_i) \end{cases} \quad (1)$$

with an after-spike resetting

$$\text{if } v \geq 30 \text{ mV, then } \begin{cases} v \rightarrow c \\ u \rightarrow u + d \end{cases} \quad (2)$$

In this description, v_i and u_i and the parameters a , b , c and d are all dimensionless and I_i is the synaptic current the neuron i receives from the other neurons.

The variable v_i is the membrane potential and u_i is the recovery variable of the neuron i . The recovery variable u indicates the activation of K^+ ionic currents and the inactivation of Na^+ ionic currents, which have a negative contribution on v_i . [1]

In this model a noise term η has been added that affects the membrane potential. Although different sources of noise exist, only one has been considered. This is the intrinsic fluctuation of the membrane potential. Typically it is represented by a random Gaussian distribution of mean 0 and SD of 3, but in this study SD will be taken as a variable (object of study) and a mean of 3 to accelerate the process and get the results in advance with fewer executions.

It is important to know that the equations of this model have a stable point. If a neuron is in the stable region it will need external inputs to get outside from this zone and be able to fire. The phase space diagram of v and u can be seen in FIG 1. This can anticipate that systems with low intensity of noise will hardly have firing neurons as the reset brings neurons to the stable zone.

Phase space diagram for the Izhikevich model

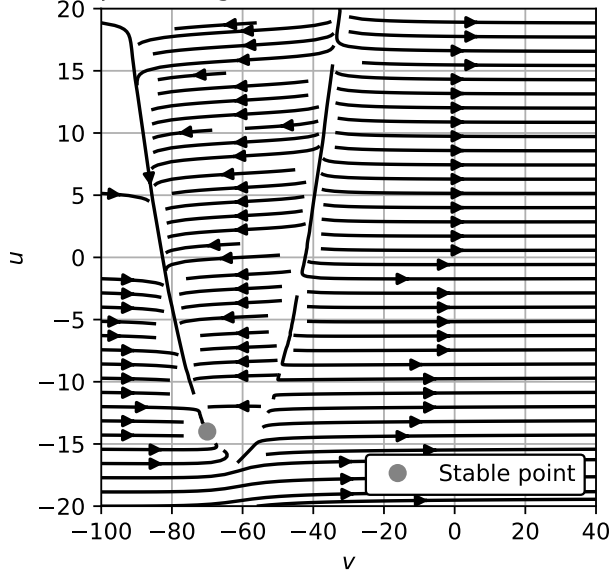


FIG. 1: The phase space diagram has a stable region inside a parabolic line, so a neuron that is inside this region will end at the stable point if it has not external inputs. The stable point is located at $(-70, -14)$. When a neuron is in the right region (unstable) it will fire if it has not a negative input and will be reset, according to Eq. 2, to the stable zone.

B. Connectivity of the system

The experimental SNN model that will be emulated consists of 4 modules disposed in a 2×2 configuration. Rather than assigning an arbitrary number of neurons to each module, it was fixed to 7 to simplify the model and the simulation. The connections of the different configurations will be done between adjacent modules and not diagonally.

The term I_i represents the contributions of the different input currents a neuron can have. In this work, the only current that will be considered is the synaptic current that becomes from the firings of neurons. The expression it takes is

$$I_i = \sum_j^N I_{ij} \quad (3)$$

$$I_{ij} = \sum_{t_l < t} W_{ij} \exp\left(-\frac{t - t_l - \Delta_{ij}}{\tau}\right) \Theta(t - t_l - \Delta_{ij}), \quad (4)$$

where the sum is done for every time that a firing takes place before a certain time, τ is the synaptic time constant set to 10 ms, Θ is the heavyside function and Δ_{ij} is a matrix that represents the synaptic delay, with a random Gaussian value for every different connection with a mean of 0.6 ms and SD of 0.1 ms.

The meaning of this expression is that every firing that has been made before a certain time t will send a synaptic current to the connected neurons. This introduces a contribution that decays exponentially over time.

In this expression $W = [W]_{ij} \in \mathbb{R}^{N \times N}$ is the weight matrix of this system. It quantifies the probability that neuron i is connected to j . This probability is determined by whether the neurons are part of the same module or not.

The value of W_{ij} will take 0 if neurons i and j are either the same neuron $i = j$ or belong to modules that are not coupled. Otherwise, if they belong to the same module and $i \neq j$ connections are formed with a probability of 1. Finally, neurons of interconnected modules are connected with a probability of $(1 + \mathcal{D})p_{int}$ for a coupling in the same direction as the connection and $(1 - \mathcal{D})p_{int}$ for the opposite. In this definition, p_{int} is the inter-module connection probability, defined in the interval $p_{int} \in [0, 0.5]$, and $\mathcal{D}$ the degree of the directionality, with bounds $\mathcal{D} \in [0, 1]$. When $\mathcal{D} = 1$, the coupling is fully oriented as the inter-module connection, and when $\mathcal{D} = 0$ it is fully random. [2] Nevertheless, in this study the value that has been taken is either 1 or 0 depending on if they are connected or not, respectively, and the number of connections between modules was fixed to 4 with an arbitrary directionality.

Besides, the sign of an element of W_{ij} is positive if the neuron j is excitatory or negative if it is inhibitory. The presence of inhibitory neurons is important to prevent big synchronization but, since the objective is based on

the impact of different topologies, in this study inhibitory neurons are not considered for simplicity.

C. Connectivity configurations

The different configurations that have been studied are named as: *feed forward*, *circular* and *sunrise*. The difference between these structures is the main direction of the connections made.

In addition, three cases have been studied for each configuration. Since there are four different neural connections between a pair of modules, the following cases have been studied: all four have the same direction (4:0), three have the main direction and 1 the opposite (3:1) and two point to one direction and the other two to the opposite (2:2). In total there are 9 cases to study but, as the configuration 2:2 is common for every system because it has not a defined direction, the number of different structures that have been analyzed is 7. The scheme of the systems can be seen in FIG. 2.

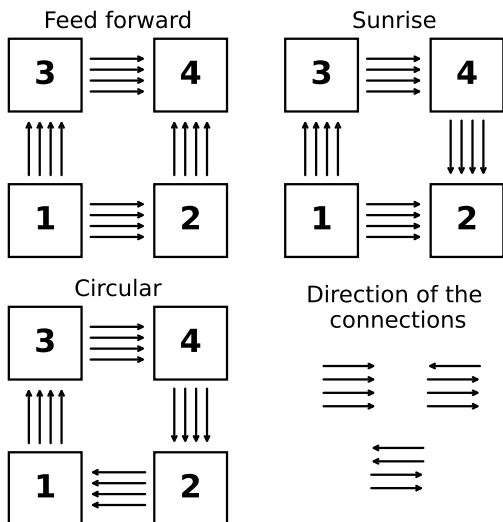


FIG. 2: Scheme of the different topologies studied. The configurations of the figure are the main structures that will be analyzed. The other ones are variations of these, changing the direction of the connections for the ones that can be seen at the bottom right (e.g. the case of the second group of arrows would be changing the direction of one connection for every group of connections of adjacent lattices).

D. Analysis of the dynamics

To determine the capacity of a certain system to replicate the major number of states an analysis of all the

firings that take place in the simulation has been done.

The result of a simulation is a vector for each neuron with the register of firings. The firing vector of the neuron i , $x_i(t)$, takes 1 at every time step that the neuron i fired and 0 when not.

The correlation of the different neurons has been calculated with the expression of the Pearson's correlation coefficients, computed as

$$r_{ij} = \frac{\sum_t [x_i(t) - \bar{x}_i][x_j(t) - \bar{x}_j]}{\sigma_i \sigma_j} \quad (5)$$

$$\sigma_i = \sqrt{\sum_t [x_i(t) - \bar{x}_i]^2}, \quad (6)$$

where $\bar{x}_i$ and σ_i are the firing rate and the standard deviation of the neuron i , respectively. If any σ_i is 0, the r_{ij} that include that neuron will automatically be set to 0.

Since there is delay on the connections of the neurons the correlation can not be seen in a single time step but in a window of time. Moreover, the first steps of the system can contain altered data as the system is still stabilizing. For this reason, before calculating the Pearson's coefficients the first time steps have been eliminated and the vector has been resized with the following rules:

1. The window size must be similar to the average of iterations between firings.
2. The resized vector will take 1 if the window contains any firing and 0 if it has not.

Upon implementing this measure, the correlation matrix r_{ij} with Pearson's coefficients can be measured properly.

To proceed, the functional complexity ϕ_{FC} has been calculated from the correlations matrix r_{ij} for a certain configuration and noise intensity with the following expression

$$\phi_{FC} = 1 - \frac{B}{2(B-1)} \sum_{k=1}^B \left| p_k(r_{ij}) - \frac{1}{B} \right|, \quad (7)$$

where B is the number of histogram bins (set to 20) and $p_k(r_{ij})$ is the probability distribution of the correlation coefficients at k -th bin (it can also be understood as the height of the k -th bin of the histogram when normalized to 1). As the correlation coefficient is high between neurons of the same module (nearly 1), the histogram and the calculation of $p_k(r_{ij})$ has only been made with the coefficients that correspond to pairs of neurons of different modules.

The functional complexity ϕ_{FC} decreases when the values of r_{ij} are localized at 0 or 1 and increases when they are widely distributed. The range of values it takes is

$\phi_{FC} \in [0, 1]$ being 0 when all the coefficients are concentrated in one bin (usually when $r_{ij} = 1 \forall i, j$ and all neurons are synchronized or $r_{ij} = 0 \forall i, j$ and the behavior is fully random) and 1 when they are uniformly distributed with $p_k(r_{ij}) = \frac{1}{B} \forall k$. This parameter can be understood as a form of entropy of the system, as it increases with the number of states the system can replicate.

III. RESULT ANALYSIS AND DISCUSSION

The simulation and the data analysis has been done all in Python (the code is available in Appendix A). The way to proceed for each system has been the following:

- The system has been simulated for a total of time of 100000 ms with a time step $dt = 1$ ms with the Euler method and this has been repeated for 81 different values of the noise SD $\sigma_\eta \in [0, 8]$.
- The parameters a , b , c and d were defined for RS neurons ($a = 0.02$, $b = 0.2$, $c = -65$ mV and $d = 8$).
- The initial conditions were: for $v_i(t = 0)$ a random value in the interval $[c, 0]$, for $u_i(t = 0)$ has been used $u_i(t = 0) = bv_i(t = 0)$ and the weight matrix W_{ij} depends on the configuration but the neurons of the connected lattices were chosen randomly.

The result of every configuration has been represented in a ϕ_{FC} vs σ_η graphic. It can be seen in FIG. 3. It can also be seen the evolution of the parameters v_i , u_i and firings vs time during the simulation and an example of histogram for the analysis at Appendix B

In order to find the peaks of the data obtained, a polynomial fit of order 5 has been done. The adjustments made can be seen in Appendix C.

To find the peaks, the point where the derivative is null has been calculated to evaluate then the adjustments at that point. The calculation of the derivatives has been done manually, not numerically, and the Brent's method with inverse quadratic interpolation [3] has been used to find the roots within an interval with the SciPy function `scipy.optimize.root_scalar` [4]. The results of the maximum of ϕ_{FC} for every group of connections can be seen in TABLE I.

$(\phi_{FC})^{max}$	Feed Forward	Circular	Sunrise
4:0	0.3405	0.1826	0.1832
3:1	0.1893	0.1729	0.1799
2:2	0.1657	0.1657	0.1657

TABLE I: Maximum of the adjustments for every configuration. As said before, 2:2 is common for all so the maximum is the same for all.

Comparing the graphics and the results of the table, is easy to identify that the configuration that is most capable of replicating the major number of states is the *feed*

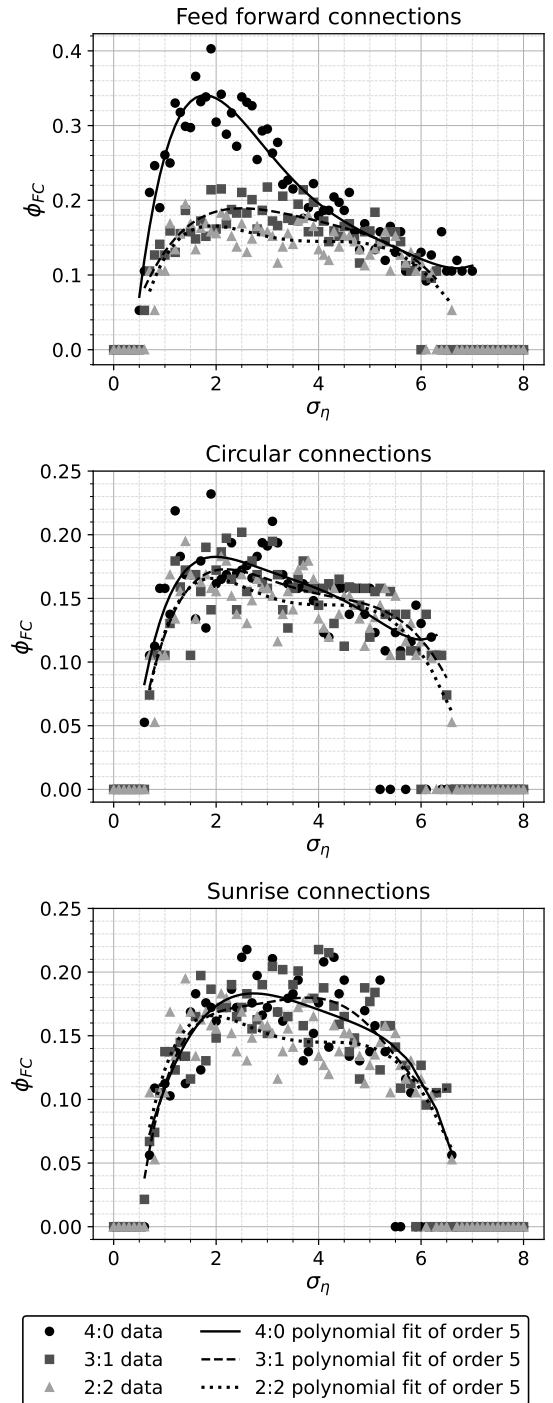


FIG. 3: Results of the configurations *feed forward*, *circular* and *sunrise* for the different type of connections (4:0, 3:1 and 2:2) in function of the standard deviation of the gaussian noise σ_η .

forward with the connections 4:0. This is reasonable because it seems that, for the direction of the connections, the module 4 depends strongly on the modules 2 and 3 and they do so with the module 1 but this one does not receive any input and is independent of the others, which

is similar to a degree of freedom.

On the other hand, although the peaks have been calculated for the other configurations, the findings remain inconclusive, as the difference between the adjustments is smaller than the statistical error that can be seen in the figures of FIG. 3.

The reason for the *sunrise* can be because, although all the directions start at 1 and end in 2, the correlation is lost due to statistical errors or noise effects as one of the paths is longer, as if the correlation were mitigated.

The result of the *circular* can be for a similar reason. As this configuration has no end or beginning the correlation mitigates on the path. It also can be because the modules have the same tendency since all of them receive contributions from other ones, being all connected.

IV. CONCLUSIONS

In this work has been found that for the different configurations studied the one that yielded the best outcome has been the *feed forward* configuration with all the connections on the same direction (4:0). This result shows the importance of the directionality of the network and is interesting because this structure of the modules and the connections seem to have a big dependence on the

first module. It might be possible to study the behavior of this configuration and the states that can replicate in function of the external input that the first module receives.

Nevertheless, in this work there is a main limitation to replicate a neural network that is able to emulate the behavior of the brain. This has been the number of neurons that are considered, with a total number of 28 neurons and 4 different modules the impact of this number in the dynamics is strong. Further works should focus on larger scale networks. Moreover, it can also be a next step to study for this model the different states that are possible to replicate in these networks.

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Diversitat d'estats en diferents esstructures de xarxes neuronals

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Resum: En aquest treball s'ha estudiat la capacitat de replicar diferents estats de diferents configuracions topològiques nombrades com *feed forward*, *sunrise* i *circular* amb diferents connexions intermodulars. Per fer-ho, s'ha utilitzat el model matemàtic d'Izhikevich per neurones de descàrrega regular excitatòries. S'han fet un conjunt de simulacions amb diferents valors de la intensitat del soroll del sistema per a un total de 7 configuracions diferents, amb tal de analitzar la evolució de la complexitat funcional ϕ_{FC} i trobar el valor del màxim amb un ajust. Finalment, s'han comparat els valors dels màxims i s'ha identificat que la millor configuració per replicar diferents estats de la xarxa neuronal ha sigut la *feed forward* amb totes les connexions intermodular en la mateixa direcció, mostrant la importància de la direccionalitat en la xarxa.

Paraules clau: Mètodes numèrics, sistemes complexos, física estadística

ODS: Salut i benestar

Objectius de Desenvolupament Sostenible (ODSs o SDGs)

1. Fi de la es desigualtats	10. Reducció de les desigualtats
2. Fam zero	11. Ciutats i comunitats sostenibles
3. Salut i benestar	X 12. Consum i producció responsables
4. Educació de qualitat	13. Acció climàtica
5. Igualtat de gènere	14. Vida submarina
6. Aigua neta i sanejament	15. Vida terrestre
7. Energia neta i sostenible	16. Pau, justícia i institucions sòlides
8. Treball digne i creixement econòmic	17. Aliança pels objectius
9. Indústria, innovació, infraestructures	

El contingut d'aquest TFG, part d'un grau universitari de Física, es relaciona amb l'ODS 3 perquè contribueix en l'estudi de certs patrons o comportaments que, com s'ha esmentat en la introducció, poden afavorir l'aparició de certes malalties i condicions neurològiques com el Parkinson, l'epilèpsia, trastorns del son o autisme entre d'altres, això passa quan les neurones del cervell estan molt sincronitzades. Tot i així, quan no hi ha sincronització, pot fins i tot arribar a ser favorable per alguns aspectes com l'atenció i la memòria.

Appendix A: Access to the code used

It has been created a GitHub repository [State-diversity-in-neural-network-models](#) where all the code used for the simulations is available.

Appendix B: Evolution of the simulation

Here it can be seen the evolution of v_i and u_i and see how the reset affects on their evolution on time. In this particular case the noise SD has a value of $\sigma_\eta \approx 1.78$ and the configuration is *feed forward* 4:0, the case where ϕ_{FC} is maximum. The evolution on time of the membrane potential $v_i(t)$ can be seen in FIG. 4, the recovery variable $u_i(t)$ in FIG. 5 and the firing times in 6.

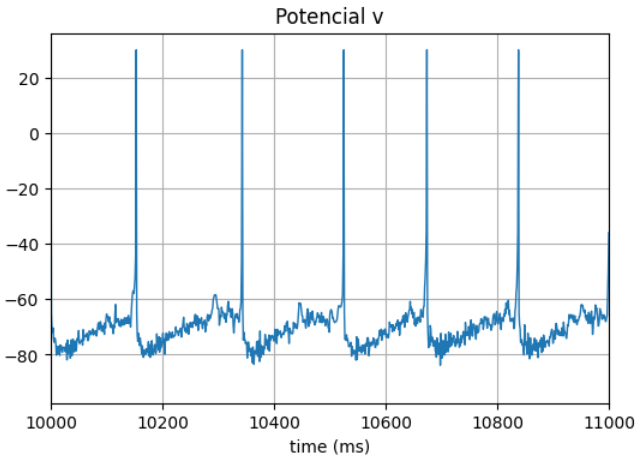


FIG. 4: Potential v_i in a window of time. The peaks are the firings that a neuron does.

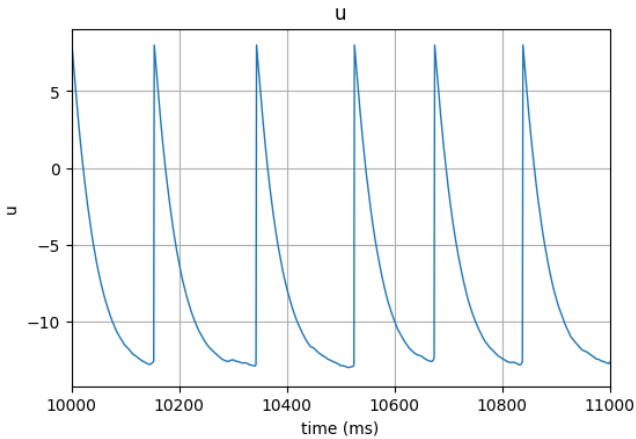


FIG. 5: Recovery variable u_i in a window of time. It can be appreciated the reset of the neurons. Comparing the values of the variable u_i with FIG. 1 it can be seen that the reset brings the neuron always to the stable zone.

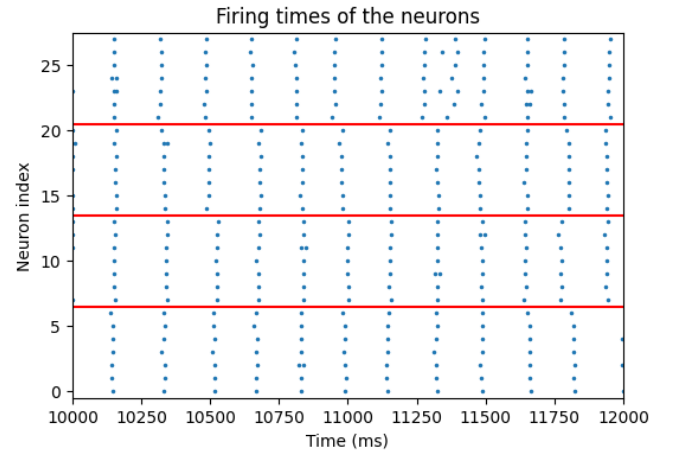


FIG. 6: Firing times of all neurons in a window of time.

For this particular case the histogram obtained has been the one shown in FIG. 7, exhibiting high function complexity, as they are not concentrated in a single box but distributed across several boxes.

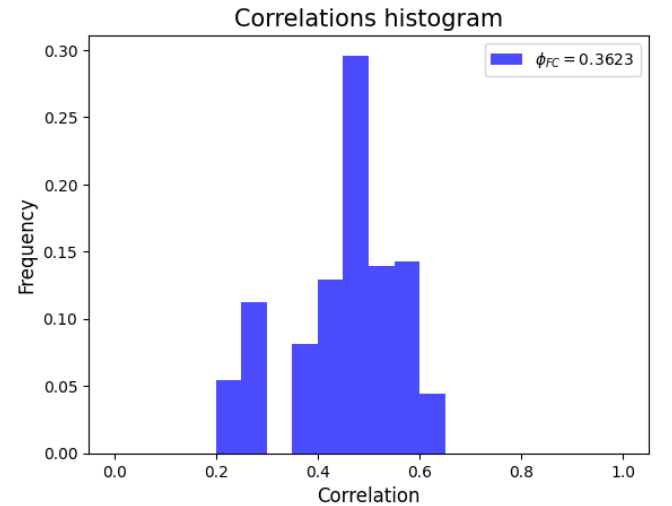


FIG. 7: Histogram of the correlation coefficients. This histogram corresponds to the case where ϕ_{FC} .

Appendix C: Adjustments

The expression of the adjustments are the following.

For *feed forward* configuration:

$$4 : 0 \rightarrow y = (-0.30 \pm 0.06) + (0.97 \pm 0.13)x^1 + \quad (C1) \\ (-0.53 \pm 0.09)x^2 + (0.12 \pm 0.03)x^3 + \\ (-0.014 \pm 0.004)x^4 + (0.0006 \pm 0.0002)x^5$$

$$3 : 1 \rightarrow y = (-0.04 \pm 0.06) + (0.25 \pm 0.14)x^1 + \quad (C2) \\ (-0.10 \pm 0.11)x^2 + (0.02 \pm 0.04)x^3 + \\ (-0.001 \pm 0.006)x^4 + (0.0000 \pm 0.0003)x^5.$$

For *sunrise* configuration:

$$4 : 0 \rightarrow y = (-0.05 \pm 0.09) + (0.22 \pm 0.19)x^1 + \quad (C3) \\ (-0.06 \pm 0.14)x^2 + (0.00 \pm 0.04)x^3 + \\ (0.001 \pm 0.007)x^4 + (-0.0001 \pm 0.0004)x^5$$

$$3 : 1 \rightarrow y = (-0.19 \pm 0.07) + (0.54 \pm 0.15)x^1 + \quad (C4) \\ (-0.33 \pm 0.11)x^2 + (0.10 \pm 0.04)x^3 + \\ (-0.014 \pm 0.005)x^4 + (0.0007 \pm 0.0003)x^5.$$

For *circular* configuration:

$$4 : 0 \rightarrow y = (-0.10 \pm 0.07) + (0.44 \pm 0.16)x^1 + \quad (C5) \\ (-0.26 \pm 0.12)x^2 + (0.07 \pm 0.04)x^3 + \\ (-0.010 \pm 0.007)x^4 + (0.0005 \pm 0.0004)x^5$$

$$3 : 1 \rightarrow y = (-0.10 \pm 0.08) + (0.36 \pm 0.16)x^1 + \quad (C6) \\ (-0.18 \pm 0.11)x^2 + (0.04 \pm 0.04)x^3 + \\ (-0.004 \pm 0.005)x^4 + (0.0001 \pm 0.0003)x^5,$$

Common for all:

$$2 : 2 \rightarrow y = (-0.14 \pm 0.07) + (0.47 \pm 0.15)x^1 + \quad (C7) \\ (-0.26 \pm 0.11)x^2 + (0.07 \pm 0.03)x^3 + \\ (-0.008 \pm 0.005)x^4 + (0.0003 \pm 0.0003)x^5,$$